

MA2041: Probability and Statistics

Problem Set 1

Discussion among friends is greatly encouraged. No need to submit. You can expect problems of this kind for quizzes and exams.

1. Complete the proof of Chebyshev's inequality.
2. The following table gives the responses of 79 students in a class when asked to write a random number between 1 to 10:

Number written	1	2	3	4	5	6	7	8	9	10
No. of students	5	7	7	9	6	8	18	8	10	1

- (a) Find the mean, 10, 25, 50, 75, 90 percentiles, and mode
 - (b) Find standard deviation.
 - (c) Draw relative frequency histogram and mark the quantities in (a) on it.
 - (d) Draw the cumulative relative frequency histogram and mark the quantities in (a) on it.
 - (e) Verify Chebyshev's result for $k = 1.5$. Compare it with the data.
 - (f) Find the inter-quartile range for the data.
3. Show that the sum of squares of deviations are minimized if deviations are taken from mean.
 4. If $r_{X,Y}$ is the correlation coefficient of the data pairs $(X_i, Y_i), i = 1, \dots, n$, and if $X'_i = a + bX_i$ for some $b < 0$, find the correlation coefficient of X'_i and Y_i . What is the significance of this result?
 5. Suppose that $(X_i, Y_i), i = 1, \dots, n$ are data pairs such that $r_{XY} = -1$. Show that there exist a and $b < 0$ such that $Y_i = a + bX_i$ for $i = 1, \dots, n$.
(Hint: When is equality in Cauchy-Schwarz inequality?)
 6. Prove that $r_{a+bX, c+dY} = \text{sign}(b) \cdot \text{sign}(d) \cdot r_{X,Y}$.
 7. We say X and Y are *uncorrelated* if $\text{Cov}(X, Y) = 0$. Show by an example that uncorrelatedness does not imply that the two variables are not related.

8. In a biology experiment a number of cultures were grown in the laboratory. The numbers of bacteria, in millions, and their ages, in days, are given below.

Age (x)	1	2	3	4	5	6	7	8
No. of bacteria (y)	37	109	140	172	205	240	280	325

Plot these values on a scatter diagram, calculate the correlation coefficient r and comment on your findings.

9. For each of the following experiments, describe the sample space.
- (i) Toss a coin four times(how many sample points).
 - (ii) The number of insect-damaged leaves on a plant.
 - (iii) Weights of rats.
 - (iv) Proportion of defectives in a shipment of electronics components.

10. Use probability axioms to prove the following:

- $A \subset B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.
- $\mathbb{P}(A) \leq 1$, for any event A .
- $\mathbb{P}(\phi) = 0$
- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$

11. Prove $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

12. If $\mathbb{P}$ is a probability function, then

- (i) For any partition $C_1, C_2, \dots$ of $\mathcal{B}$ (σ -algebra on Ω),

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A \cap C_i).$$

$C_1, C_2, \dots$ is called a a partition of $\mathcal{B}$ if $C_i \cap C_j = \phi$ whenever $i \neq j$, and $\mathcal{B} = \cup_{i=1}^{\infty} C_i$.

- (ii) Define $A_1^* = A_1$, $A_i^* = A_i \setminus (\cup_{j=1}^{i-1} A_j)$, $i \geq 2$. Show that $A_i^* \cap A_k^* = \phi$ whenever $i \neq k$.

- (iii) Next show that $\sum_{i=1}^{\infty} \mathbb{P}(A_i^*) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

- (iv) Can you conclude *Boole's inequality* now?
13. Prove the generalised *Bonferoni's inequality*: $A_1, A_2, \dots, A_n \in \mathcal{B}$, then
- $$\mathbb{P}(\cap_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbb{P}(A_i) - (n-1).$$
14. 3 coins are tossed. Find the probability that exactly and atleast one of the coins is a head.
15. 2 coins, one with $\mathbb{P}(\text{head}) = u$ and one with $\mathbb{P}(\text{head}) = w$, are to be tossed together independently. Define

$$\begin{aligned} p_0 &= \mathbb{P}(0 \text{ heads occur}), \\ p_1 &= \mathbb{P}(1 \text{ head occurs}), \\ p_2 &= \mathbb{P}(2 \text{ heads occur}). \end{aligned}$$

Can u and w be chosen such that $p_0 = p_1 = p_2$? Justify your answer.

16. In the game of darts (discussed in class), where there are 1,2,3,4,5 points to be scored, with 5 rings labelled 5 in the innermost circle descending to 1 on the outermost ring, find the probability of scoring i points. What kind of a function (monotone or not) is this with respect to i ?
17. If $\mathbb{P}(A) = \frac{1}{3}$ and $\mathbb{P}(B^c) = \frac{1}{4}$, can A and B be disjoint?
18. Try all problems of Chapters 1 & 2 of Sheldon Ross' book.

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Problem Set 2

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. If n balls are placed at random into n cells, what is the probability that exactly one cell remains empty?
2. (i) Show that the number of ways to partition n distinct objects into k distinct groups containing $n_1, n_2, \dots, n_k$ objects, respectively (i.e. $n_1 + n_2 + \dots + n_k = n$), is

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

- (ii) What is the probability that a 8 letter word made with the letters from the word PALAKKAD starts with K ?
3. Two dice are thrown. Find the probability that the sum of the faces equals or exceeds 8.
4. Suppose that a bag contains 100 slips such that in 95 slips the letter 'A' is written and in 5 slips letter 'B' is written. Three slips are drawn at random with replacement. What is the probability of getting two or more A's?
5. 3 fair dice are rolled, where every die has the numbers $[1,7]$. Find the probability that the sum would be 14.
6. Suppose that a fair coin is tossed until a head is obtained.
 - (i) Describe the sample space.
 - (ii) What is the probability that exactly k tosses are required?
 - (iii) What is the probability that k is odd?
7. A closet contains n pairs of shoes. If $2r(< n)$ shoes are chosen at random, what is the probability that there will be no matching pair in the sample?
8. Compute the probability that 10 married couples are seated at random at a round table, then no wife sits next to her husband.

9. The Fisher's have two children. At least one of them is a boy. What is the probability that both children are boys?(For a detailed discussion on this problem see *Gardner 1961*.)
10. Let B be an event such that $P(B) > 0$. Show that $P(\cdot|B)$ satisfies the axioms of probability.
11. Prove the following.
 - (a) A and B are independent iff $P(A \cap B) = P(A)P(B)$
 - (b) if A and B are independent, then A and B^c are also independent.
12. A collection of subsets $A_1, \dots, A_n \subseteq S$ is said to be *mutually independent* if for every subset $\{i_1, \dots, i_r\} \subseteq \{1, \dots, n\}$ we have

$$P(A_{i_1} \cap \dots \cap A_{i_r}) = P(A_{i_1}) \cdots P(A_{i_r}).$$

Show that for n mutually independent events $A_1, \dots, A_n$,

$$P(A_1 \cup \dots \cup A_n) = 1 - (1 - P(A_1))(1 - P(A_2)) \cdots (1 - P(A_n)).$$

Hint: Use induction on n .

13. A collection of subsets $A_1, \dots, A_n \subseteq S$ is said to be *pairwise independent* if $P(A_i \cap A_j) = P(A_i)P(A_j)$ for all $i \neq j$.
Give an example to show that pairwise independent does not always imply mutually independent.
14. There are n types of coupons, and each new one collected is independently of type i with probability p_i , $\sum_{i=1}^n p_i = 1$. Suppose k coupons are to be collected. If A_i is the event that there is at least one type i coupon among those collected, then find
 - (i) $\mathbb{P}(A_i)$,
 - (ii) $\mathbb{P}(A_i \cup A_j)$, for $i \neq j$,
 - (iii) $\mathbb{P}(A_i|A_j)$, for $i \neq j$.

15. Assume that there is a test for covid that is 95% accurate. Suppose that an individual takes this test and tests positive. Also assume that the person comes from a population where 2% of them suffer from covid. What is the probability that the person really has covid?
16. Cards are dealt one at a time from a standard pack. First 2 cards are both spades. What is the probability that the next 3 cards are spades?
17. Jacky is a Manchester City fan and likes to drink black coffee whenever his team loose, and prefers masala tea otherwise. Jacky has two separate bags containing coffee and tea sachets. Bag 1 contains 8 masala tea sachets and 7 coffee sachets, Bag 2 contains 3 masala tea sachets and 12 coffee sachets. Jacky reaches for Bag 2 when Man city loses and reaches for Bag 1 otherwise, and without looking picks up a sachet at random. Suppose after Manchester derby (a match between Man City and Man United), Jacky ends up drinking masala tea. What is the probability that Man City lost in that derby?
18. An ordinary deck of 52 cards is randomly divided into 4 piles with 13 cards each. Compute the probability that each pile has exactly 1 ace.
19. Urn 1 has n red balls and urn 2 has n blue balls. Balls are randomly removed from urn 1 in the following manner: after each removal from urn 1, a ball is taken from urn 2 (if urn 2 has any balls), and placed in urn 1. The process continues until all the balls have been removed (so there are total $2n$ removals).
 - (i) Find the probability that the final ball removed from urn 1 is red.
 - (ii) Repeat the problem when urn 1 has r_1 red balls and b_1 blue balls, and urn 2 has r_2 red balls and b_2 blue balls.
20. A student is taking a one-hour-time-limit makeup exam. Suppose the probability that the student will finish in less than x hours is $x/2$, for all $0 \leq x \leq 1$. Then, given that the student is still working after 0.75 hour, what is the conditional probability that the full hour is used?
21. Independent trials consisting of rolling a pair of dice is performed. What is the probability that an outcome of 5 appears before an outcome of 7 when the outcome of a roll is the sum of the dice?

22. A and B play a series of games. Each game is independently won by A with probability p and by B with probability $(1 - p)$. They stop when the total number of wins of one of the players is two greater than that of the other player. The player with the greater number of total wins is declared the winner.
- (i) Find the probability that a total of 4 games are played.
 - (ii) Find the probability that A is the winner in the series.
23. (*Birthday Problem*) If n people are present in a room, what is the probability that no two of them celebrate their birthday on the same day of the year? How large need n be so that this probability is less than $\frac{1}{2}$?
24. Consider a *complete graph* with n vertices. There are $\binom{n}{2}$ edges connecting each pair of vertices. Now suppose, each edge in the graph is to be coloured either red or blue. For a fixed integer k , is there a way of colouring the edges such that no set of k vertices has all of its $\binom{k}{2}$ connecting edges the same colour? (*Hint*: Let E_i denote the event that the i th set of k vertices are of same colour, for $1 \leq i \leq \binom{n}{k}$, and use Boole's inequality to get a bound on n .)
25. Try as many problems of Chapter 3 of Sheldon Ross as you can.

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Problem Set 3

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Let X have the uniform distribution on the first N natural numbers, i.e.,

$$\mathbb{P}[X = k] = \frac{1}{N}, \text{ for all } 1 \leq k \leq N.$$

Find the expected value and variance of X .

2. Let X be a random variable with pdf

$$f(x) = \begin{cases} \frac{2}{x^3}, & x \geq 1, \\ 0, & x < 1. \end{cases}$$

What can you say about the expectation and variance of X ?

3. The distribution function(cdf) of the random variable X is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{2}, & 0 \leq x < 1, \\ \frac{2}{3}, & 1 \leq x < 2, \\ \frac{11}{12}, & 2 \leq x < 3, \\ 1, & 3 \leq x. \end{cases}$$

Compute

- (a) $\mathbb{P}(X < 3)$,
(b) $\mathbb{P}(X > \frac{1}{2})$,
(c) $\mathbb{P}(2 < x \leq 4)$.
4. (a) (*Hypergeometric distribution*) A *hypergeometric* random variable X with parameters n, N, m represents the number of white balls selected when n balls are chosen at random from an urn that contains N balls of which m are white. The pmf of X is given as follows:

$$\mathbb{P}(X = i) = \frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}}, \quad i = 0, 1, 2, \dots, m.$$

Find $\mathbb{E}[X]$ and $\text{Var}(X)$.

- (b) The hypergeometric distribution has application in *acceptance sampling*. For instance, suppose, a retailer buys goods in lots and each item can be either acceptable or defective. Let N be the total items in a lot and m be the number of defectives in a lot. Then we can calculate the probability that a sample of size n contains i many defective items via the random variable X as described in part (a).

Suppose that a lot of 25 machine parts is delivered, where a part is acceptable only if it passes tolerance. We sample 10 parts and find that none are defective(all within tolerance). What is the probability of this event if there are 6 defectives in the lot of 25?

5. Find the mean and variance of the negative binomial distribution with parameters r, p . Y is a negative binomial variable with pmf

$$\mathbb{P}(Y = y) = \binom{r + y - 1}{y} p^r (1 - p)^y, \quad y = 0, 1, 2, \dots$$

What happens to the mean and variance if $r \rightarrow \infty$ and $p \rightarrow 1$ such that $r(1 - p) \rightarrow \lambda, 0 < \lambda < \infty$?

6. A *truncated* discrete distribution is one in which a particular class cannot be observed and is eliminated from the sample space. In particular if X takes values $0, 1, 2, \dots$ and the class 0 cannot be observed, the *0-truncated* random variable, X_T has pmf

$$\mathbb{P}(X_T = x) = \frac{\mathbb{P}(X = x)}{\mathbb{P}(X > 0)}, \quad x = 1, 2, \dots$$

Find the mean and variance of the 0-truncated random variable starting from

- (a) $X \sim \text{Poisson}(\lambda)$,
- (b) $X \sim \text{negative binomial}(r, p)$.

7. Median of a random variable X is defined as a number m such that $P(X \leq m) \geq \frac{1}{2}$ and $P(X \geq m) \geq \frac{1}{2}$.

- (a) Does a median always exist for a random variable?

- (b) Suppose X is a random variable with distribution $P(X = 1) = 1/4, P(X = 2) = 1/2, P(X = 3) = 1/4$. Realize the danger in defining the median to be an m such that $F_X(m) = 1/2$.
 - (c) Show, however, that for a continuous random variable X , median is the value m such that $F_X(m) = 1/2$.
- 8. A certain typing agency employs 2 typists. The average number of errors per article is 3 when typed by the first typist and 4.2 when typed by the second. If your article is equally likely to be typed by either typist, approximate the probability that it will have no errors.
- 9. People enter a gambling casino at a rate of 1 every 2 minutes. What is the probability that
 - (a) no one enters between 12:00 and 12:05?
 - (b) at least 4 people enter the casino between 12:00 and 12:05?
- 10. A family has n children with probability $\alpha p^n, n \geq 1$, where $\alpha \leq \frac{1-p}{p}$.
 - (a) What proportion of families has no children?
 - (b) If each child is equally likely to be a boy or a girl (independently of each other), what proportion of families consists of k boys (and any number of girls)?
- 11. Buses arrive at a specified stop at 15-minute intervals starting at 7 am. That is they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits
 - (a) less than 5 minutes for a bus,
 - (b) more than 10 minutes for a bus.
- 12. Let X be uniformly distributed over $(0, 1)$. Define a random variable Y as $Y = X^n$. Find the pdf of Y .

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Problem Set 4

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Let Φ denote the CDF of a normal random variable $X \sim N(\mu, \sigma^2)$, that is,

$$\Phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt.$$

Show that $\Phi(-x) = 1 - \Phi(x)$ for $x \geq 0$.

2. If $Z \sim N(0, 1)$, find the distribution of Z^2 . Recognize that this is a gamma distribution.
3. Verify that for any $\alpha > 0$, $\Gamma(\alpha + 1) = \Gamma(\alpha)$.
4. Define the *Beta function* as

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx, \quad \text{for } \alpha, \beta > 0.$$

Now define

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1, \alpha, \beta > 0.$$

Then prove the following:

- (a) $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.
- (b) $f(x)$ is a pdf.
- (c) Find the mean and variance of the probability distribution.
5. Define $F(x) = \begin{cases} 1 - e^{-x^2}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$
- (a) Verify that $F(x)$ is a cdf for a random variable X .
- (b) Find $\mathbb{P}(X > 2), \mathbb{P}(1 < X < 3)$.
- (c) Find $\mathbb{E}[X]$.
6. Suppose $X \sim \text{Exp}(\lambda)$ with $f_X(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}, x > 0, \lambda > 0$. Then find the pdf of the following:
- (a) $Y = X^{\frac{1}{\gamma}}, \gamma > 0$.

- (b) $Y = (2X/\lambda)^{1/2}$.
 - (c) With $\lambda = 1$, $Y = \log(X)$.
7. Suppose $X \sim \text{Normal}(0, 1)$, then find the distribution of e^X .
 8. Suppose $X \sim \text{Normal}(10, 36)$, then find
 - (a) $\mathbb{P}(X > 5)$.
 - (b) $\mathbb{P}(4 < X < 16)$.
 - (c) $\mathbb{P}(X < 20)$.
 9. Suppose that X is a normal random variable with mean 5. If $\mathbb{P}(X > 9) = 0.2$, approximately what is $\text{Var}(X)$.
 10. Let $X \sim \text{Normal}(12, 4)$. Find the value of c such that $\mathbb{P}(X > c) = 0.10$.
 11. Suppose that height(in inches) of a 25-year-old man is a normal random variable with parameters, $\mu = 71$ and $\sigma^2 = 6.25$. What percentage of 25-year-old men are over 6 feet tall?
 12. A standard *Cauchy random variable* has pdf

$$f(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < \infty.$$

Show that if X is a standard Cauchy variable then $1/X$ is also a standard Cauchy random variable.

13. Find the first four moments of a $N(\mu, \sigma^2)$ random variable.
14. Find the standard deviation of a random variable X whose moment generating function is given by $\phi_X(t) = \frac{1}{(2-e^t)^3}$.
15. Find the MGF for a Binomial random variable with parameters n, p and for a Poisson random variable with parameter λ . The use those to verify their expectation and variance.
16. A random variable X is said to be *exponentially distributed* with parameter λ , that is, $X \sim \text{Exp}(\lambda)$, if it has the following pdf.

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0. \end{cases}$$

Find its MGF and hence its mean and variance.

17. A random variable X with the following pdf

$$f(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} x^{\alpha-1} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

is called a *gamma* random variable with parameters $\alpha > 0, \lambda > 0$. Show that its MGF is $(\frac{\lambda}{\lambda-t})^\alpha$.

18. Let $f(x)$ be a pdf and let a be a number such that, for all $\epsilon > 0$, $f(a+\epsilon) = f(a-\epsilon)$. Such a pdf is said to be symmetric about the point a .

- (a) Give three examples of symmetric pdfs.
- (b) Show that if $X \sim f(x)$, symmetric and $\mathbb{E}[X]$ exists, then $\mathbb{E}[X] = a$.
- (c) Is the χ_n^2 variable (chi-squared random variable with n degrees of freedom) symmetric about 0?
 - If X is a random variable with distribution symmetric about 0, then show that its MGF $M_X(t)$ is symmetric (as a function in t) about 0.

19. Let $M_X(t)$ be the MGF of X and define $C(t) = \log(M_X(t))$. Show that

$$\left. \frac{d}{dt} S(t) \right|_{t=0} = \mathbb{E}[X], \quad \text{and} \quad \left. \frac{d^2}{dt^2} S(t) \right|_{t=0} = \text{Var}(X).$$

20. Try as many problems as you can from Chapter 5 of Sheldon Ross' book.

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Problem Set 5

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Show that, if $F_{X,Y}$ is differentiable at (a, b) , then

$$\frac{\partial^2}{\partial x \partial y} F(x, y)|_{(a,b)} = f_{X,Y}(a, b).$$

2. Prove that

$$P(a_1 < X \leq a_2, b_1 < Y \leq b_2) = F(a_1, b_1) + F(a_2, b_2) - F(a_1, b_2) - F(a_2, b_1).$$

3. Show that the random variables X and Y (whether discrete or continuous) are independent iff $F_{X,Y}(x, y) = F_X(x)F_Y(y)$.
4. Suppose X and Y are independent random variables. Find the distribution of $Z = X + Y$, when the distribution of X and Y are given as follows:

- (a) $X \sim \text{Poi}(\lambda_1)$, $Y \sim \text{Poi}(\gamma)$,
- (b) $X, Y \sim \text{Geometric}(p)$,
- (c) $X, Y \sim \text{Exp}(\lambda)$,
- (d) $X \sim \text{Gamma}(\alpha, \beta)$, $Y \sim \text{Gamma}(\lambda, \beta)$.

5. Show that uncorrelated does not, in general, imply independence.
6. Show, however, if $X, Y \in \{0, 1\}$, then uncorrelated implies independence. How do you interpret this?
(Hint: The passage in p.126 of Sheldon Ross' book concerning indicator random variables will be helpful)
7. Markov's inequality of order p : Show that if X is a non-negative random variable with $\mathbb{E}[X^p] < \infty$, then $P(X > a) \leq \mathbb{E}[X^p]/a^p$ for any $a > 0$ (Hint: Split the expectation). What is the significance of this result?
8. The correlation between two random variables X and Y is ρ_{XY} . If, $\rho_{XY} = 1$, what is the relation between X and Y ?
9. A random point (X, Y) is distributed uniformly on the square with vertices $(1, 1)$, $(1, -1)$, $(-1, 1)$ and $(-1, -1)$, i.e., the joint pdf is $f(x, y) = \frac{1}{4}$ on the aforementioned square. Calculate $\mathbb{P}(X^2 + Y^2 < 1)$, $\mathbb{P}(2X - Y > 0)$, $\mathbb{P}(|X + Y| < 2)$.

10. Suppose the joint pdf of (X, Y) is given as

$$f(x, y) = \begin{cases} 2x, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{P}(X^2 < Y < X)$.

11. Let X and Y be independent geometric variables with parameter p .
- (a) Find the joint pmf of X and $X + Y$.
 - (b) Define $U = \min(X, Y)$ and $V = X - Y$. Show that U and V are independent.
12. Let X and Y are independent standard normal variables. Find the pdf of $\frac{(X-Y)^2}{2}$.
13. A point is generated at random in the plane according to the following polar scheme. A radius R is chosen, where the distribution of R^2 is χ^2 with 2 degrees of freedom. Independently, an angle θ is chosen, where $\theta \sim \text{Unif}(0, 2\pi)$. Find the joint distribution of $X = R \cos(\theta)$ and $Y = R \sin(\theta)$.
14. Show that any random variable is uncorrelated with a constant.
15. Suppose $(X, Y) \sim \text{bivariate normal}(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho)$. Then prove the following:
- (a) The marginal distribution of X is normal(μ_x, σ_x^2).
 - (b) The marginal distribution of Y is normal(μ_y, σ_y^2).
 - (c) For constants a and b , find the distribution of $aX + bY$.
16. Let $X \sim \text{normal}(\mu, \sigma^2)$ and $Y \sim \text{normal}(0, \sigma^2)$. Suppose X and Y are independent. Define $U = X + Y$ and $V = X - Y$. Find the joint distribution of (U, V) . Are U and V independent? What is the distribution of U ?
17. Let X and Y be independent $\text{Uniform}(0, 1)$ random variables. Find $\mathbb{P}(X/Y \leq t)$ and $\mathbb{P}(XY \leq t)$.

18. Let X_1, X_2 and X_3 be uncorrelated random variables, each with mean μ and variance σ^2 . Find $\text{cov}(X_1 + X_2, X_2 + X_3)$ and $\text{cov}(X_1 + X_2, X_1 - X_2)$.
19. Find the pdf of $\prod_{i=1}^n X_i$, where the X_i s are independent $Unif(0, 1)$ random variables.
20. A parallel system is one that functions as long as at least one component of it functions. A particular system is composed of three independent components, each of which has lifelength with an $Exponential(\lambda)$ distribution. The lifetime of the system is the maximum of the individual lifelengths. What is the distribution of the lifetime of the system?
21. Let X and Y be two random variables. Use *Jensen's inequality* to prove the following (*Holder's inequality*):

$$\mathbb{E}[|XY|] \leq (\mathbb{E}[|X|^p])^{\frac{1}{p}} (\mathbb{E}[|Y|^q])^{\frac{1}{q}},$$

where $\frac{1}{p} + \frac{1}{q} = 1$.

22. Solve as many problems as possible from Chapter 4 of Sheldon Ross' book.

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Problem Set 5

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Let $X_1, X_2, \dots, X_n$ be i.i.d. (independent and identically distributed) with continuous cdf F and mean μ . Define random variable $Y_1, Y_2, \dots, Y_n$ as

$$Y_i = \begin{cases} 1, & \text{if } X_i > \mu, \\ 0, & \text{if } X_i \leq \mu. \end{cases}$$

Find the distribution of $\sum_{i=1}^n Y_i$.

2. Let X and Y be i.i.d. (independent and identically distributed) standard normal variables. Define $Z = \min(X, Y)$. Show that $Z_1^2 \sim \chi_1^2$.
3. Let $X_1, X_2, \dots$ be a sequence of positive random variables that converges in probability to a constant a . Prove that $Y_i = \sqrt{X_i}, i \geq 1$ converges in probability to $\sqrt{a}$ and $Z_i = \frac{a}{X_i}$ converges in probability to 1.
4. Does convergence in distribution imply convergence in probability? Justify.
5. (*Poisson Approximation to Binomial*) Suppose $X \sim \text{Bin}(n, p)$. Show that, when n is “large” and p is “small”,

$$\Pr(X = i) \approx \frac{e^{-\lambda} \lambda^i}{i!},$$

where $\lambda = np$.

6. (Galton Board) Let $X_1, \dots, X_n$ be i.i.d. with distribution $\Pr(X_i = -1) = \Pr(X_i = 1) = 1/2$. Show that $X_1 + \dots + X_n \stackrel{d}{\approx} \mathcal{N}(0, n)$.
7. For a random variable X with MGF $M(t)$, prove the following inequality:

$$\mathbb{P}(X \geq a) \leq e^{-ta} M(t), \quad t > 0.$$

8. (a) Suppose X and Y are independent Poisson random variables with parameters λ and μ respectively. Show that $X + Y$ is also a Poisson random variable with parameter $\lambda + \mu$. Give two different proofs (i)

without using the Moment Generating Functions and (ii) using the Moment Generating Functions.

(b)(*Normal approximation to Poisson*) Show by Central Limit Theorem that, if X_n is $\text{Poi}(n)$, then $X_n \stackrel{d}{\approx} N(n, n)$. (Hint: Show that a $\text{Poi}(n)$ random variable is a sum of n independent $\text{Poi}(1)$ random variables (first show it for two variables)).

9. Let $X_1, X_2, \dots, X_{10}$ be i.i.d. $\text{Uniform}(0, 1)$ variables. Calculate an approximation to $\mathbb{P}(X_1 + X_2 + \dots + X_{10} > 6)$.
10. An instructor has 50 exams that will be graded in a sequence. The times required to grade the 50 exams are independent, with a common distribution that has mean 20 minutes and standard deviation 4 minutes. Approximate the probability that the instructor will grade at least 25 of the exams in the first 450 minutes of work.
11. A company manufactures batteries. Each lifetime (in hours) of a battery is assumed to follow a probability distribution with mean 100 and standard deviation 20. What minimum sample size will be required to test to conclude with at least 95 percent certainty that the average lifetime of a battery lies between 95 to 105 hours.
12. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with variance σ^2 . Find $\text{cov}(X_i - \bar{X}, \bar{X})$.
13. Suppose the daily number of customers in a new cafe follows a Poisson distribution with mean 5. A new cafe owner has been keeping track of all customers that have visited her cafe until the footfall count exceeds 100. That is if Y_n represents the number of customers visited till the n th day, the $Y_n = Y_{n-1} + X_n, n \geq 1$, where $X_1, X_2, \dots$ are i.i.d. $\text{Poi}(5)$ variables. What can you say about the probability that she will be able to stop counting after 15 days?
14. Problems 1 – 17 of Chapter 6 of Sheldon Ross' book.

MA2041: Probability and Statistics

Problem Set 7

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Let $X_1, X_2, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find a function of the sample variance S^2 , say $g(S^2)$ that satisfies $\mathbb{E}[g(S^2)] = \sigma$. (*Hint*: Try $g(S^2) = c\sqrt{S^2}$, where c is a constant to be determined).
2. Let $X_i, i = 1, 2, 3$, be independent with $N(i, i^2)$ distributions. For each of the following, use X_i s to construct a statistic(function of $X_i, i = 1, 2, 3$) with the indicated distribution:
 - (a) χ_3^2 , chi squared with 3 degrees of freedom
 - (b) t_2 , t distribution with 2 degrees of freedom
 - (c) $F_{1,2}$, F distribution with 1 and 2 degrees of freedom.
3. Let X be a random variable with a Student's t distribution with p degrees of freedom.
 - (a) The distribution of X is symmetric, i.e., $X \stackrel{d}{=} -X$.
 - (b) Find the mean and variance of X .(when do they exist)
 - (c) Find the distribution of X^2 . Identify this distribution.
4. Let X be a random variable with an $F_{p,q}$ distribution. Then
 - (a) find $\mathbb{E}[X]$ (when does it exist)
 - (b) find $\text{Var}(X)$ (when does it exist)
 - (c) show that $\frac{1}{X}$ has an $F_{q,p}$ distribution
 - (d) show that $\frac{\frac{p}{q}X}{1+\frac{p}{q}X}$ has a Beta distribution with parameters $\frac{p}{2}$ and $\frac{q}{2}$.
5. Given an i.i.d. sample $X_1, \dots, X_n$ from the underlying model, find the MLE of the unknown parameter in each of the following cases:
 - (a) Geometric distribution: $\text{Geo}(p)$.
 - (b) Truncated exponential:

$$f(x|\theta) = \begin{cases} e^{-(x-\theta)} & \text{for } x > \theta, \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta \in \mathbb{R}$.

(c) Laplace distribution:

$$f(x|\theta) = \frac{1}{2}e^{-|x-\theta|}, \quad -\infty < x, \theta < \infty,$$

6. Suppose that $X_1, \dots, X_n$ is an i.i.d. sample drawn according to a particular member of

$$f(x|\theta) = \begin{cases} e^{-(x-\theta)} & \text{for } x > \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Verify if MLE of θ is unbiased.

7. Let $X_1, X_2, \dots, X_n$ be a random sample from a $\text{Gamma}(\alpha, \beta)$ population. Find the MLE of β , assuming α is known.
8. Let $X_1, X_2, \dots, X_n$ be a random sample from the pdf

$$f(x|\theta) = \theta x^{-2}, \quad 0 < \theta \leq x < \infty.$$

Find the MLE of θ .

9. Determine the MLE of a and λ when $X_1, X_2, \dots, X_n$ is a random sample from the Pareto density

$$f(x|a, \lambda) = \begin{cases} \lambda a^\lambda x^{-(\lambda+1)}, & \text{if } x \geq a, \\ 0, & \text{if } x < a. \end{cases}$$

10. Problems 18–30 of Chapter 6 of Sheldon Ross' book (solve as many as possible)

MA2041: Probability and Statistics

Problem Set 8

Note: Discussion among friends is greatly encouraged. But write solutions on your own. No need to submit.

1. Let $X_1, X_2, \dots, X_n$ be a random sample drawn from a population having distribution $N(\mu, 4)$. For testing the hypotheses

$$H_0 : \mu = 2 \text{ versus } H_1 : \mu = 3,$$

a test with critical region

$$C = \{(X_1, X_2, \dots, X_n) | \bar{X} > 2.5 \text{ or } \bar{X} < 1.5\}$$

is given. Then

- (a) Find the probability of Type-I and II errors.
- (b) Find the power of the test.

2. Consider a trial in which a jury must decide between the hypothesis that the defendant is guilty and the hypothesis that he or she is innocent.

- (a) In the framework of hypothesis testing and the India legal system, which of the hypotheses should be the null hypothesis?
- (b) What do you think would be an appropriate significance level in this situation?

3. A sugar refiner packs sugar into bags with an average weight of 1kg. Unfortunately, the filling equipment tends to drift - it is known that the process standard deviation is 0.1. From a sample of 100 bags the sample mean was 1.03. Is the process mean still 1kg? Use a 0.05 level of significance.

4. In a certain chemical process, it is very important that a particular solution that is to be used as a reactant have a pH of exactly 8.20. A method for determining pH that is available for solutions of this type is known to give measurements that are normally distributed with a mean equal to the actual pH and with a standard deviation of .02. Suppose 10 independent measurements yielded the following pH values:
8.18, 8.17, 8.16, 8.15, 8.17, 8.21, 8.22, 8.16, 8.19, 8.18

- (a) What conclusion can be drawn at the $\alpha = .10$ level of significance?
- (b) What about at the $\alpha = .05$ level of significance?

5. There is some variability in the amount of phenobarbital in each capsule sold by a manufacturer. However, the manufacturer claims that the mean value is 20.0 mg. To test this, a sample of 25 pills yielded a sample mean of 19.7 with a sample standard deviation of 1.3. What inference would you draw from these data? In particular, are the data strong enough evidence to discredit the claim of the manufacturer? Use the 5 percent level of significance.
6. The mean response time of a species of pigs to a stimulus is .8 seconds. Twenty eight pigs were given 2 ounce of alcohol and then tested. If their average response time was 1.0 seconds with a standard deviation of .3 seconds, can we conclude that alcohol affects the mean response time? Use the 5 percent level of significance.
7. An advertisement for a new toothpaste claims that it reduces cavities of children in their cavity-prone years. Cavities per year for this age group are normal with mean 3 and standard deviation 1. A study of 2,500 children who used this toothpaste found an average of 2.95 cavities per child. Assume that the standard deviation of the number of cavities of a child using this new toothpaste remains equal to 1.
- (a) Are these data strong enough, at the 5 percent level of significance, to establish the claim of the toothpaste advertisement?
 - (b) Do the data convince you to switch to this new toothpaste?
8. The weights of salmon grown at a commercial hatchery are normally distributed with a standard deviation of 1.2 pounds. The hatchery claims that the mean weight of this year's crop is at least 7.6 pounds. Suppose a random sample of 16 fish yielded an average weight of 7.2 pounds. Is this strong enough evidence to reject the hatchery's claims at the
- (a) 5 percent level of significance;
 - (b) 1 percent level of significance?
 - (c) What is the p-value?
9. Last year, 456 students of IIT Palakkad had given the CAT examination. These students had an average score of 60 with a standard deviation of 5.6. The national average was 56.5. Test the claim of an

administrator that IIT Palakkad students scored significantly higher than the national average. Let $\alpha = 0.05$.

10. Consider a test of $H_0 : \mu \leq 100$ versus $H_1 : \mu > 100$. Suppose that a sample of size 20 has a sample mean of $\bar{X} = 105$. Determine the p-value of this outcome if the population standard deviation is known to equal 5.
11. A medical scientist believes that the average basal temperature of (outwardly) healthy individuals has increased over time and is now greater than 98.6 degrees Fahrenheit (37 degrees Celsius). To prove this, she has randomly selected 100 healthy individuals. If their mean temperature is 98.74 with a sample standard deviation of 1.1 degrees, does this prove her claim at the 5 percent level? What about at the 1 percent level?
12. An oil company claims that the sulfur content of its diesel fuel is at most .15 percent. To check this claim, the sulfur contents of 40 randomly chosen samples were determined; the resulting sample mean and sample standard deviation were .162 and .040. Using the 5 percent level of significance, can we conclude that the company's claims are invalid?
13. A manufacturer of capacitors claims that the breakdown voltage of these capacitors has a mean value of at least 100 V. A test of 12 of these capacitors yielded the following breakdown voltages:
96, 98, 105, 92, 111, 114, 99, 103, 95, 101, 106, 97
Do these results prove the manufacturer's claim? Do they disprove them?
14. A car is advertised as having a gas mileage rating of at least 30 miles/gallon in highway driving. If the miles per gallon obtained in 10 independent experiments are
26, 24, 20, 25, 27, 25, 28, 30, 26, 33
Should you believe the advertisement? What assumptions are you making?
15. It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of current gain is 200 with a sample standard deviation of 35, would the claim be rejected at the 5 percent level of significance if

- (a) the sample size is 25?
 - (b) the sample size is 64?
16. Certain rockets are manufactured with a range of 2,500 meters. It is theorized that the range will be reduced after the rockets are in storage for some time. Six of these rockets are stored for a certain period of time and then tested. The ranges found in the tests are as follows: 2,490, 2,510, 2,360, 2,410, 2,300, and 2,440. Does the range appear to be shorter after storage? Test at the 5% significance level. Here, the variation in ranges is also of importance. New rockets have a standard deviation of range measurements equal to 20 kilometers. Does it appear that storage increases the variability of these ranges? Use $\alpha = 0.05$.
 17. A machined engine part produced by a certain company is claimed to have a diameter variance no larger than 0.0002 inch. A random sample of ten parts gave a sample variance of 0.0003. Assuming normality of diameter measurements, is there significant evidence to refute the company's claim? Use $\alpha = 0.05$.
 18. A gun-like apparatus has recently been designed to replace needles in administering vaccines. The apparatus can be set to inject different amounts of the serum, but because of random fluctuations the actual amount injected is normally distributed with a mean equal to the setting and with an unknown variance σ^2 . It has been decided that the apparatus would be too dangerous to use if σ exceeds .10. If a random sample of 50 injections resulted in a sample standard deviation of .08, should use of the new apparatus be discontinued? Suppose the level of significance is $\alpha = .10$. Comment on the appropriate choice of a significance level for this problem, as well as the appropriate choice of the null hypothesis.
 19. Aptitude tests should produce scores with a large amount of variation so that an administrator can distinguish between persons with low aptitude and persons with high aptitude. The standard test used by a certain industry has been producing scores with a standard deviation of 5 points. A new test is tried on 20 prospective employees and produces a sample standard deviation of 8 points. Are scores from the

new test significantly more variable than scores from the standard?
Use $\alpha = 0.05$.

20. A small business has 37 employees. Because of the uncertain demand for its product, the company usually pays overtime per week, whatever is required, and the variance on this figure is about 25. Company officials want to know whether the variance of overtimes hours has changed. Assume hours of overtime are normally distributed, and $\alpha = 0.05$. Given here is a sample of 16 weeks of overtime data (in hours per week).

57, 56, 52, 44, 46, 53, 44, 44, 48, 51, 55, 48, 63, 53, 51, 50.

21. A manufacturer claims that the lifetime of a certain brand of batteries produced by his factory has a variance of 5000 (hours)². A sample of size 26 has a variance of 7200 (hours)². Assuming that it is reasonable to treat these data as a random sample from a normal population. Let us test the manufacturer's claim at the 0.02 level.
22. A pharmaceutical house produces a certain drug item whose weight follows normal distribution with mean 5.7 and standard deviation 0.5 milligrams. The company's research team has proposed a new method of producing the drug. However, this entails some costs and will be adopted only if there is strong evidence that the standard deviation of the weight of the items will drop to below 0.5 milligrams. If a sample of 10 items is produced and has the following weights, should the new method be adopted? Let $\alpha = 0.05$.

5.728, 5.731, 5.722, 5.719, 5.727, 5.724, 5.726, 5.718, 5.723, 5.722.

23. The U.S. Farmers' Production company builds large harvesters. For a harvester to be properly balanced when operating, a 25 pound plate is installed on its side. The machine that produces these plates is set to yield plates that average 25 pounds weight. The distribution of plates produced from the machine is normal. However, the shop supervisor is worried that the machine is out of adjustment and is producing plates that do not average 25 pounds. To test his concern, he randomly selects 20 of the plates produced the day before and weights them. Below shows the weights obtained.

22.6, 22.2, 23.2, 27.4, 24.5, 27, 26.6, 28.1, 26.9, 24.9, 26.2, 25.3, 23.1, 24.2,

- 26.1, 25.8, 30.4, 28.6, 23.5, 23.6. Do the test at 5% level of significance.
24. It is known that if a signal value μ is sent from location A , then the value received at location B is normally distributed with mean μ and standard deviation 2. That is, the random noise added to the signal is an $N(0, 4)$ random variable. There is reason for the people at location B to suspect that the signal value $\mu = 8$ will be sent today. Test this hypothesis at 10% level of significance, if the same signal value is independently sent five times and the average value received at location B is 9.5.
25. The mean breaking strength of a certain type of fibre is required to be at least 200 psi. Past experience indicates that the standard deviation of breaking strength is 5 psi. If a sample of 8 pieces of fibre yielded breakage at the following pressure: 210, 195, 197, 199, 198, 202, 196, 195. Would you conclude, at the five percent level of significance that the fibre is unacceptable? What about at the 10% level of significance?